

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator
- **Domain:** HealthTech & Artificial Intelligence

2. THE CONTEXT OF PERSONAL FITNESS

Achieving fitness goals requires a combination of structured workouts and proper nutrition. However, human bodies are fundamentally different. A workout routine that yields excellent results for a 20-year-old athlete might be ineffective or even dangerous for a 40-year-old beginner. Personalization is the key to sustainable fitness.

3. THE FITNESS BOTTLENECK

Currently, individuals looking to start their fitness journey face two major hurdles:

- **The Information Overload:** Searching the internet for workout plans yields thousands of generic, conflicting results. Beginners often do not know which exercises suit their specific age, weight, and fitness goals.
- **The Financial Barrier:** Hiring a certified personal trainer or a professional nutritionist to create a customized daily plan is expensive and not accessible to everyone.

4. THE CORE PROBLEM STATEMENT

"Fitness enthusiasts and beginners lack a free, accessible, and highly personalized tool that can instantly generate a structured, day-by-day workout routine tailored strictly to their unique physical metrics and personal goals."

5. INADEQUACY OF EXISTING SOLUTIONS

Most standard fitness applications operate on rigid, pre-programmed templates. They ask for a user's BMI and assign a generic "Weight Loss" or "Muscle Gain" PDF. They lack the dynamic ability to adapt to nuanced user preferences, such as specific intensity levels or unique physical constraints.

6. THE FITBUDDY SOLUTION FitBuddy addresses this problem by acting as an AI-powered virtual personal trainer. By leveraging Google Gemini Generative AI, the platform dynamically analyzes the user's Age, Weight, Fitness Goal, and preferred Workout Intensity. It then instantly generates a completely customized 7-day workout plan and nutrition tip, ensuring the routine is perfectly aligned with the individual's capabilities.